
[All ETDs from UAB](#)

[UAB Theses & Dissertations](#)

2022

The Effect of Federally Qualified Health Centers (FQHCS) Prevalence on Uncompensated Care in US Hospitals

Kena R. Norris

University Of Alabama At Birmingham

Follow this and additional works at: <https://digitalcommons.library.uab.edu/etd-collection>



Part of the [Medical Humanities Commons](#)

Recommended Citation

Norris, Kena R., "The Effect of Federally Qualified Health Centers (FQHCS) Prevalence on Uncompensated Care in US Hospitals" (2022). *All ETDs from UAB*. 144.

<https://digitalcommons.library.uab.edu/etd-collection/144>

This content has been accepted for inclusion by an authorized administrator of the UAB Digital Commons, and is provided as a free open access item. All inquiries regarding this item or the UAB Digital Commons should be directed to the [UAB Libraries Office of Scholarly Communication](#).

THE EFFECT OF FEDERALLY QUALIFIED HEALTH CENTERS (FQHCs)
PREVALENCE ON UNCOMPENSATED CARE IN U.S. HOSPITALS

by

KENA R. NORRIS

ROBERT HERNANDEZ, COMMITTEE CHAIR

NATHANIEL CARROLL

AMY YARBROUGH LANDRY

WILLIAM OPOKU-AGYEMAN

A DISSERTATION

Submitted to graduate faculty of The University of Alabama at Birmingham,
in partial fulfillment of the requirements for the degree of
Doctor of Science

BIRMINGHAM, ALABAMA

2022

Copyright by
Kena R. Norris
2022

THE EFFECT OF FEDERALLY QUALIFIED HEALTH CENTERS (FQHCs) PREVALENCE ON UNCOMPENSATED CARE IN U.S. HOSPITALS

KENA R. NORRIS

ADMINISTRATION – HEALTH SERVICES

ABSTRACT

Over the past two decades, uncompensated care in U.S. hospitals has grown at a staggering rate and reached \$46.4 billion in 2013. It began to decline in 2014, due to the passage of the Patient Protection and Affordable Care Act (ACA), but is now on the rise again. Uncompensated care is typically affected by policy and market factors such as legislative changes and an increase in the number of uninsured. While hospitals continue to serve as critical safety net providers, they are often not the only or most appropriate facility to serve the indigent in U.S. communities. Instead, federally qualified community health centers (FQHCs) were designed to provide care for the poor and uninsured.

Accordingly, studies have shown that FQHCs reduce the number of costly emergency department visits and preventable hospitalizations. However, little work has been done showing the effect of FQHC penetration on hospitals' uncompensated care in a shared geographic area.

In this study, FQHCs were posited as the health policy of interest to determine the effect of FQHC presence on uncompensated care in U.S. hospitals. Structural equation modeling was employed to compare uncompensated care in counties with FQHCs versus counties without an FQHC. In the final model, FQHCs, Medicaid

expansion, race, high school education, uninsured population, household, unemployment rate, ED visits, and HMO penetration were statistically significant.

The findings in this study revealed that counties with FQHC presence had higher uncompensated care. Further research is needed to understand why FQHCs' presence increases uncompensated hospital care. Moreover, a better understanding of healthcare utilization, such as patient visits and referral patterns, between FQHCs and hospitals may help to further inform researchers about this finding.

Keywords: Uncompensated care, federally qualified health centers (FQHCs), community health centers, Medicaid expansion

ACKNOWLEDGMENTS

First and foremost, I want to thank God for making this possible. See Mark 10:27. Despite the challenges and hardships, He made a way for me when there seemed to be no way.

I would also like to thank my mother, Sandra, who started this doctorate journey with me, but was called to her Heavenly home. “Ma” helped to instill in me the confidence that I needed to succeed, and she always provided loving support.

I am grateful for the insight my chair, Dr. Robert “Bob” Hernandez imparted. When I faced my mom's unexpected loss, followed by my own cancer diagnosis, Bob shared with me that “life will eventually get out of the way.” I needed to hear that perspective, so I could eventually finish my dissertation.

A warm thanks to my committee members, Dr. William Opoku-Agyeman, Dr. Nathaniel Carroll, and Dr. Amy Landry, for their patience and support that also helped to propel me forward.

Much gratitude to Ms. Leandra Y. Celaya for her impeccable operational support for the DSc program and its students. Thank you to Dr. Larry R. Herald for his initial guidance and confidence in my project.

A big ‘shout out’ to the eighth cohort, which was such a unique cast of characters. I will always remember you fondly because my time in the program would not have been the same without you.

And thank you to all my family and friends who encouraged me along the way.

TABLE OF CONTENTS

	Page
ABSTRACT.....	iii
ACKNOWLEDGEMENTS.....	v
LIST OF TABLES.....	viii
LIST OF FIGURES.....	ix
CHAPTER ONE: INTRODUCTION.....	1
Background.....	1
Purpose of the Study.....	3
Research Questions.....	4
CHAPTER TWO: LITERATURE REVIEW.....	5
Uncompensated Care.....	5
Medicaid Expansion.....	7
Federally Qualified Health Centers (FQHC).....	9
Effect of FQHCs Presence on Hospitals.....	11
Theoretical Framework.....	14
Organizational Ecology.....	14
Ecology of Medical Care.....	15
Access to Medical Care.....	18
Conceptual Model.....	21
Research Hypotheses.....	22
CHAPTER THREE: METHODOLOGY.....	24
Research Methods.....	24
Study Measures.....	24
Dependent Variable.....	24
Independent Variable.....	25
Control Variables.....	25
Statistical Analysis.....	29
CHAPTER FOUR: RESULTS.....	32

CHAPTER FIVE: DISCUSSION.....	45
Review and Assessment of Findings.....	45
Limitations.....	47
Recommendations for Future Research.....	49
Conclusion.....	50
REFERENCES.....	51
APPENDIX.....	57
A CORRELATION MATRIX.....	57
B STRUCTURAL EQUATION MODELING (SEM).....	59
C SEM WITH MEDICAID EXPANSION.....	61
D SEM WITH MEDICAID EXPANSION AS MODERATOR.....	63
E IRB APPROVAL.....	65

LIST OF TABLES

<i>Table</i>	<i>Page</i>
1. Summary of Study Variables.....	27
2. Summary of Geographic Distribution.....	30
3. Descriptive Statistics for National Data.....	33
4. Analysis of Hypothesis 1- FQHCs Prevalence to Uncompensated Care...	34
5. Rural Analysis.....	36
6. Micro Metro Analysis.....	37
7. Metropolitan Analysis.....	38
8. Descriptive Statistics for California and Illinois.....	39
9. California and Illinois Analysis.....	40
10. Analysis of Hypothesis 2 – Effect of Medicaid Expansion.....	41
11. Medicaid Expansion as a Moderator Analysis.....	42
12. By Year Analysis, 2015-2019.....	43
13. Correlation Matrix.....	57

LIST OF FIGURES

<i>Table</i>	<i>Page</i>
1. Conceptual Model.....	21
2. Structural equation modeling (SEM).....	59
3. SEM with Medicaid Expansion.....	61
4. SEM with Medicaid Expansion as Moderator.....	63

CHAPTER 1

INTRODUCTION

Background

Millions of uninsured Americans utilize healthcare services, but cannot afford to pay their medical bills. It was estimated that in 2013, the cost of uncompensated care delivered to the uninsured was \$84.9 billion, but the majority (60%) of their care was provided by hospitals (Coughlin et al., 2014). Over the past two decades, uncompensated care in U.S. hospitals has grown at a staggering rate and reached \$46.4 billion in 2013 (AHA Uncompensated Care, 2017). According to the American Hospital Association's Annual Survey of Hospitals, uncompensated care began to decline in 2014, mainly due to Medicaid expansion under the Patient Protection and Affordable Care Act (ACA) but is now on the rise again. Thus, U.S. hospitals provided \$38.3 billion in uncompensated care in 2016, up from \$35.7 billion in 2015 (AHA Uncompensated Care, 2017).

Uncompensated care, comprised of charity care and bad debt, is typically affected by policy and market factors like legislative changes, an increase in the number of uninsured, and financial pressures from payors (Bazzoli et al., 2006). While the ACA held great potential to reduce uncompensated care, there will always be several Americans who remain uninsured even after its full implementation, particularly those living in states that did not expand Medicaid, undocumented immigrants, and those who cannot find affordable coverage or choose to remain uninsured (Coughlin et al., 2014).

Moreover, according to the Medicaid and CHIP Payment and Access Commission (MACPAC), the decline in uncompensated care was more significant in the first year of ACA implementation (MACPAC, 2018). Charity care and bad debt, as a share of hospital operating expenses, fell 18 percent from 2013 to 2014 compared to 14 percent from 2014 to 2015. Thus, the year-over-year effects of coverage expansion of the ACA appear to have a diminishing impact on hospitals' uncompensated care (MACPAC, 2018). This will be further exacerbated by the rising rate of uninsured, up 1.3 percentage points to 12.2% in 2017 from 2016, according to the Gallup-Sharecare Well-Being Index, representing an increase of approximately 3.2 million Americans (Auter, 2018). Meanwhile, the cost of uncompensated care burdens hospitals by making it more challenging to maintain the capacity to serve patients, invest in newer technologies, or remain financially solvent to keep their doors open (Center on Budget and Policy Priorities, 2018).

In addition, rising competition, the need for cost containment, and expanding market share of for-profit hospitals raise the question about the supply of safety net providers (Banks et al., 1997). Nevertheless, hospitals continue to serve as critical safety net providers. However, they are often not the only, or the most appropriate, facility to serve the indigent in many U.S. communities. Instead, community health centers were designed to provide care for the poor and uninsured and must-see patients regardless of their ability to pay. Although federally qualified health centers (FQHCs) have existed since 1965, these organizations are often overlooked healthcare providers, but perceptions are changing.

Studies have shown that FQHCs reduce the number of costly emergency department visits and ambulatory-sensitive inpatient admissions (Epstein, 2001; Falik et al., 2001; Probst et al., 2009; Richard et al., 2012). Due to their ability to provide

broad reductions in healthcare utilization and spending, FQHCs have garnered substantial bipartisan support for expansion and doubled the capacity of patients served under President Bush from 9.6 million in 2000 to 19.5 million by 2009 (Richard et al., 2012). Subsequently, the ACA provided an additional \$11 billion in mandatory funding from 2011 to 2015 to further multiply the capacity of FQHCs in the coming years (Richard et al., 2012). According to the National Association of Community Health Centers website, over 1,400 FQHCs serve more than 27 million primarily uninsured or underinsured patients in over 9,800 rural and urban areas nationwide (NACHC, 2018).

A study found that safety net providers are “interdependent,” and hospitals experience a more significant burden caring for the indigent whenever other safety net providers cannot meet the needs of a community (Trzeciak, 2003). However, with the expansion of FQHCs in recent years, there are more facilities to care for the poor and uninsured than ever. Hospitals with an FQHC within their service area should experience a reduced burden to serve these patient populations, decreasing uncompensated care.

Purpose of the Study

Studies have shown that FQHCs positively affect healthcare utilization and spending by reducing emergency department visits and preventable hospitalizations (Epstein, 2001; Falik et al., 2001; Probst et al., 2009; Richard et al., 2012). Moreover, The Commonwealth Fund issued the following recommendation, “Develop policies to encourage health centers and local hospitals to formalize partnerships in order to assume shared accountability for low-income and uninsured patients” to improve care coordination and management of illness for the poor and uninsured (Doty et al., 2010).

However, there is minimal literature on the impact of FQHC penetration on hospitals, particularly regarding economic benefits. This study will provide an ecological analysis of FQHCs, posited as the health policy of interest, to determine if improved access to medical care via community health centers affect uncompensated care at area hospitals. Using data from the American Hospital Association (AHA), Medicare Cost Reports, Uniform Data System (USD), and Area Health Resources Files (AHRF) from 2015-2019, this study will assess the impact of FQHC prevalence on uncompensated care for U.S. hospitals.

Research Questions

In developing the questions for this study, it was clear that there was a lack of information on the fiscal impact of community health centers on area hospitals. It was suggested that further research was needed to quantify the economic benefit of community health centers in reducing ED visits by the uninsured (Rust et al., 2009). Due to the low-income and uninsured populations that community health centers serve, it seemed logical that a hospital's rate of uncompensated care would be affected by FQHC presence. Thus, this study will examine the economic benefit of community health centers by analyzing the effects of FQHC presence on the level of uncompensated care in U.S. hospitals.

1. Does access to medical care at FQHCs affect uncompensated care provided in U.S. hospitals?
2. Does Medicaid expansion affect the relationship between FQHC presence and uncompensated care in U.S. hospitals?

CHAPTER 2

LITERATURE REVIEW

This chapter begins with a broad evolution of uncompensated care, Medicaid expansion, and federally qualified health centers (FQHCs). A literature review explored the relationship between FQHCs and U.S. hospitals. It is followed by theoretical frameworks and resulting hypotheses on the effect of FQHC presence on uncompensated care in U.S. hospitals and Medicaid expansion as a moderating factor.

Uncompensated Care

In many communities, there is limited access to healthcare services for the poor and uninsured. Prior to the implementation of the Patient Protection and Affordable Care Act (ACA), hospital emergency departments (EDs) experienced a striking 36% increase in patient volumes from 1996 to 2006 (Lacalle, 2010). The rising number of uninsured Americans and increasing patient volumes in EDs were not coincidental, particularly with the enactment of the Emergency Medical Treatment and Labor Act (EMTALA). Subsequently, hospital EDs have served as the primary safety-net provider for the uninsured in many communities across the United States (Asplin, 2001). The EMTALA forced hospitals to stabilize and treat anyone presenting in the ED, regardless of their insurance status or ability to pay. However, it remains an unfunded mandate since its enactment in 1986 (Vickery et al., 2013).

Moreover, hospital EDs serve as a point of entry to inpatient care. According to Schuur et al. (2012), hospital admissions grew by 15 percent from 1993 to 2006, with a 50.4 percent increase and 11.5 million to 17.3 million admissions coming from the ED. Furthermore, all inpatient stays from the ED rose from 33.5 to 43.8 percent.

Although the ED is critical to hospital operations, many administrators view the department as a “loss leader” due to its financial burden, partly due to the uncompensated care provision required by the law (Morganti et al., 2013).

According to the 2017 American Hospital Association (AHA) factsheet on uncompensated care, hospitals have provided over \$576 billion in uncompensated care since 2000. Uncompensated care comprises bad debt and charity care and serves as the overall measure of hospital care for which no payment was received by patients or insurers. Charity care provides financial assistance to low-income, uninsured patients for medical services, but the hospital does not expect payment. Bad debt is incurred when the hospital expects payment but cannot obtain reimbursement for the care provided. However, it is essential to note that uncompensated care excludes unfunded care costs, such as underpayment from Medicaid or Medicare.

Hospitals have a long tradition of providing charity and discounted care to indigent patients, which results in uncompensated care (Bazzoli et al., 2006). Depending on the hospital’s mission, ownership status, geographical location, financial position, and other factors, the hospital will budget for some level of charity care. Patients who cannot afford to pay are identified and classified as either charity care or bad debt. It is challenging for hospitals to distinguish between the two because bad debt is often generated by low-income and/or uninsured patients (AHA Uncompensated Care, 2017). Moreover, every hospital does it differently, so uncompensated care is measured by the sum total of bad debt and charity care (Hsieh et al., 2010).

Many factors impact the level of uncompensated care delivered in U.S. hospitals, but several studies have shown that the number of uninsured, health policy changes, and financial pressures from payers have a significant effect (Mann et al.,

1997; Bazzoli et al., 2006; Hsieh et al., 2010). Thus, it stands to reason that the ACA would considerably impact uncompensated care for hospitals as it initially required states to expand Medicaid coverage to low-income populations.

Medicaid expansion under the ACA has significantly impacted uncompensated care in U.S. hospitals due to those coverage gains. It initially required states to expand Medicaid coverage to low-income populations with household incomes up to 138% of the federal poverty level. This was later ruled unconstitutional, and state participation became optional. According to The Henry J. Kaiser Family Foundation, as of July 2018, 34 states, including Washington D.C., have adopted some form of Medicaid expansion. Subsequently, over 73.5 individuals enrolled in Medicaid and CHIP as of May 2018, nearly 16 million more individuals and a 28 percent increase from enrollment prior to Medicaid expansion (Medicaid & CHIP Enrollment, 2018). Studies have found that uncompensated care in U.S. hospitals has been significantly reduced by Medicaid expansion (Dranove et al., 2016; Camilleri, 2018).

Medicaid Expansion

Medicaid expansion could decrease unreimbursed expenses attributable to the uninsured while increasing revenue from the newly covered, reducing uncompensated care (Blavin, 2016). Medicaid expansion under the ACA has significantly impacted uncompensated care in U.S. hospitals due to those coverage gains. An early study on Medicaid expansion examined the impact on Connecticut, a state that immediately expanded its Medicaid program after the passage of the ACA. Using data from 2007-2013, it found that in contrast to national and regional uncompensated care trends, Connecticut had not increased in uncompensated care and was about a one-third lower than it would have been without early Medicaid expansion (Nikpay et al., 2015).

A national study examining the impact of Medicaid expansion on hospital financing from 2011-2014 found similar results. The findings showed that hospitals in expansion states had a \$2.8 million decrease in uncompensated care and a \$3.2 million increase in Medicaid revenue relative to hospitals in non-expansion states (Blavin, 2016). Studies using more recent data have found similar results, indicating that uncompensated care in U.S. hospitals has been significantly impacted by Medicaid expansion (Dranove et al., 2016; Camilleri, 2017).

Moreover, Medicaid expansion has had other positive effects beyond the reduction of uncompensated care. According to Searing and Hoadley (2016), the resulting financial improvements have allowed hospitals to invest in hiring new staff, opening clinics, and buying new equipment. Furthermore, the authors found more integration of care for hospitals and federally qualified health centers through new systems and relationships in Medicaid expansion states, which may also be remarkably beneficial in delivering care to the remaining 28.2 million uninsured Americans (CDC, 2017).

Despite the initial successes of the ACA in reducing uncompensated care, it remains an issue for many U.S. hospitals. Moreover, the ACA legislation included reductions for disproportionate share hospital (DSH) payments and Medicaid and Medicare allotments to cover uncompensated care, which are now scheduled for implementation in 2020. However, the national amount of DSH allotment funding in FY 2020 (\$15.7 billion) is less than the amount of hospital uncompensated care (\$44.9 billion) in 2015, so safety-net hospitals may continue to face financial challenges due to uncompensated care (MACPAC, 2018).

Until the ACA, uncompensated care had remained relatively stable at about six percent of hospital expenses nationally (Hadley et al., 2008; Coughlin et al.,

2014). However, a study of Pennsylvania hospitals showed that the indirect cost of uncompensated care was also responsible for decreasing compensated care by two percent on average for hospitals (Ferrier et al., 2006). Uncompensated care requires using resources that could have been utilized to provide compensated care. This “crowding out” of compensated care represents an indirect opportunity cost to hospitals, which can affect a hospital’s ability to obtain debt financing due to reduced potential earnings. Thus, hospitals with higher-than-average levels of uncompensated care could negatively impact profitability and jeopardize their ability to carry out their missions (Ferrier et al., 2006).

In the past, hospitals could fill the gaps in the U.S. healthcare coverage system by charging more to insured patients to treat uninsured patients in a practice known as cost-shifting (Mann et al., 1995). However, as Medicare, Medicaid, and other payors introduced fixed fee payments, the financial risks were shifted from payors to hospitals, reducing their ability to cost-shift (Ferrier et al., 2006). Further, market pressures drove some hospitals and physicians to focus intensely on activities that generate revenue, including expanding into more affluent neighborhoods, downsizing or eliminating unprofitable services, and reducing budgeted funds for indigent care (Cunningham et al., 2008). Thus, these changes in strategy may increase the need for other safety-net providers, such as FQHCs, to provide healthcare access for the poor and insured in those communities.

Federally Qualified Health Centers

In areas with an established community health center, those patient populations can access primary care and preventative health services regardless of their ability to pay. Studies have shown that patients receiving medical care at community health centers have lower healthcare utilization and spending due to fewer emergency

department visits and preventable hospitalizations (Epstein, 2001; Falik et al., 2001; Probst et al., 2009; Richard et al., 2012), which could result in reduced levels of uncompensated care for hospitals in the same service area.

According to the National Association of Community Health Centers (NACHC), community health centers were born out of community health and civil rights activism to improve healthcare access and outcomes for poor Americans in need of healthcare. While studying in South Africa, Dr. H. Jack Geiger noticed the remarkable health improvements the community health center model brought about for the country's most impoverished citizens (NACHC, 2018). Dr. Geiger then sought to capitalize on Lyndon B. Johnson's "War on Poverty" initiatives in the early 1960s, so he and other healthcare activists submitted proposals to the federal Office of Economic and the first two community health centers – Boston, Massachusetts, and Mound Bayou, Mississippi – were funded in 1965, which launched the community health center program (NACHC, 2018).

Community health centers were designed to address not only the medical needs of the patients but also the socio-economic barriers to care. Many earlier centers provided food assistance, job training, housing, legal assistance, and transportation to address the other factors that impact health (Hesketh & Dwiel, 2017). It was the vision of the initial activists that community health centers would address more than just the physical health of low-income and underserved populations (Jedele, 2016). Now over 50 years later, most federally qualified health centers (FQHCs), as renamed in 1990, still provide a holistic approach to medical care for those populations. Furthermore, the services provided at a health center must be comprehensive and culturally competent for all of its patients, regardless of income, race, education, or identity.

As recipients of grant funding by Section 330 of the Public Health Services (PHS) Act (42 U.S.C. §254b), community health centers are uniquely qualified to serve low-income and uninsured patients. Section 330, from the Health Center Consolidation Act of 1996, consolidated all the health center categories (community health, migrant health centers, homeless health, and public housing health centers) under one designation so that all of the qualifying health centers receiving federal funding were referred to “Federally Qualified Health Centers” (Hesketh & Dwiel, 2017). In addition, FQHC “look-a-likes” receive funding from the state, as opposed to federal funding, but must adhere to the exact requirements, which, per the PHS Act, are as follows:

- Be located in a federally designated medically underserved area (MUA) or serve a federally designated medically underserved population (MUP)
- Have non-profit or public status.
- Provide services to patients regardless of their ability to pay; provide a full discount to patients at or below 100% of the federal poverty level guidelines and a sliding scale fee for incomes up to 200% of the poverty level.
- The majority of the governing board must be patients at the health center and be representative of its demographics, such as race, ethnicity, and gender.
- Provide comprehensive primary care services, referrals, and other services needed to facilitate access to care, such as case management, transportation, and translation services.

Effect of FQHCs Presence on Hospitals

Community health centers serve a comparatively challenging and complex patient population compared to most medical practices. FQHC patients are more likely to be uninsured, Medicaid enrollees, racial/ethnic minorities, and rural dwellers,

and have higher rates of chronic diseases (Forrest & Whalen, 2000; Laiteerapong et al., 2014). However, despite these challenges, several studies have shown that patients who receive care at FQHCs have lower emergency department visits (Smith-Campbell, 2005; Rust et al., 2009) and fewer preventable hospitalizations (Epstein, 2001; Probst et al., 2009; Rothkopf et al., 2011).

Moreover, a study looking at health center penetration, defined by the number of FQHC patients as a share of the total number of low-income residents in a hospital referral region, and the association between Medicare spending and clinical quality found that higher FQHC penetration resulted in lower utilization rates and costs for Medicare beneficiaries (Sharma et al., 2014). Likewise, a study comparing utilization by Medicaid patients seen at FQHCs to private providers found that community health center users were a third less likely to have emergency department visits, inpatient hospitalizations, and preventable hospital admissions (Rothkopf et al., 2011).

A study by Laiteerapong et al. (2014) confirmed the reduction of hospitalizations but also found that FQHC patients had fewer office visits than other primary care sites. However, FQHC patients still received more preventative care than non-FQHC patients. The authors suggested this was the result of comprehensive care, as required by the FQHC mission, and the more significant uncertainty that these patients will follow up so thorough care is provided at each visit.

However, most notably, Laiteerapong et al. (2014) found lower utilization rates and similar or more preventative care were also provided for uninsured patients at community health centers. The authors suggested that the focus on comprehensive care by FQHCs may prevent further complications or increased severity of illnesses for this vulnerable population, reducing the need for recurrent outpatient visits and hospitalizations. Nevertheless, FQHCs remain a unique healthcare provider as they

may be the only access point of care for the uninsured. Without FQHCs, many of these patients would end up in the emergency department or be hospitalized with preventable health conditions, resulting in uncompensated care at U.S. hospitals.

Finally, a study by Bazzoli et al. (2006) looking at market factors that influenced uncompensated care at urban general acute care hospitals from 1996-2000 found that FQHCs had a significant negative effect on uncompensated care for non-safety net hospitals, categorized as defined by their analyses, but was not the case for core safety-net and voluntary safety-net hospitals. The authors posited that because FQHCs deliver primary care, greater FQHC capacity may allow non-safety net hospitals to reduce uncompensated primary care services in their emergency rooms. Although FQHCs were the control variable, it was the only study, to the best of our knowledge, that included FQHCs as a measure when examining uncompensated care. However, this study had some acknowledged limitations as it was explicitly designed to explore the effects of the Balanced Budget Act (BBA) of 1997 on uncompensated care. Moreover, it was restricted to metropolitan statistical areas, which limits its generalizability for all U.S. hospitals.

Although there have been a few studies around the effect of FQHCs on ED visits or preventable hospitalization rates (Epstein, 2001; Falik et al, 2001; Probst et al, 2009; Richard et al, 2012), there has only been one study that tangentially looked at FQHCs in relationship to uncompensated care at hospitals (Bazzoli et al., 2006). In addition, it was conducted with data mainly from the 1990s, which may have little relevance to the impact of FQHCs today. As previously stated, due to substantial federal policy changes, FQHCs have grown exponentially in the number of delivery sites, and patients served since 2000. Hence, this study seeks to address some of the

limitations of prior studies and examine the present-day effect of FQHCs on uncompensated care by U.S. hospitals.

Theoretical Framework

Organizational Ecology

The proliferation of FQHCs was no coincidence and can best be explained by organizational ecology, as it provides insight into how social, economic, and political environmental conditions impact organizational diversity and changing composition over time (Baum & Shipilov, 2006). Organizational ecology proposes that the environment selects the optimal composition of organizations (Hannan & Freeman, 1977).

The natural selection model posits that environmental factors will select organizational characteristics that best fit the environment (Aldrich & Pfeffer, 1976). Natural selection applies to the population level of organizations and implies that those organizations are moving to a better environmental fit (Aldrich & Pfeffer, 1976). Hannan & Freeman (1977) suggested that organizational ecology should seek to understand the distribution of organizations across environmental conditions using a population of organizations with a shared fate as the unit of analysis and a system relevant to the study of organization-environment relations, which is usually defined by geography, political boundaries, market or product considerations.

Typically, the model addresses the rates of organizational findings, deaths, and changes in organizational forms with a focus on the selection process (Singh & Lumsden, 1990). Organizations with the proper structure, for whatever the basis, are selected over those that do not (Aldrich & Pfeffer, 1976). In its classic application, organizational ecology states that once organizations are founded, they are subject to

strong inertial pressures, so changes in organizational populations are mostly comprised of organizational foundings and dissolutions (Singh & Lumsden, 1990).

Accordingly, it is no surprise that the propagation of FQHCs coincided with the rise in uncompensated care as their organizational structure makes them uniquely positioned to serve the uninsured. Over the prior two decades, uncompensated care in U.S. hospitals grew at a staggering rate and reached \$46.4 billion in 2013 (AHA Uncompensated Care, 2017). Although the health center program started in 1965, federal initiatives such as the Health Center Growth Initiative in 2002 and the \$11 billion Community Health Center Fund under the Affordable Care Act in 2010 have greatly expanded the growth of FQHCs (Chang et al., 2019). In comparing FQHC users to non-users, health centers provided significantly lower costs and generated \$24 billion in savings with FQHC patients served in 2009 (Ku et al., 2009). The financial investments in FQHCs were primarily prompted by federal policymakers seeking a more efficient and cost-effective healthcare delivery system.

In keeping with the organizational ecology model, social, economic, and political conditions precipitated the rapid expansion of FQHCs as the environment selects the optimal composition of organizations (Baum & Amburgey, 2000; Hannan & Freeman, 1977). Subsequently, FQHCs grew by over 80 percent between 2007 and 2014 (Chang et al., 2019). In 2014, FQHCs served 23 million patients. 28 percent were uninsured, making them an important part of the healthcare safety net (Weiss et al., 2016).

Ecology of Medical Care

Given that safety net providers are interdependent (Trzeciak, 2003), it would stand to reason that a community health center's availability (or lack thereof) could impact the frequency of care at a local hospital in the same geographical area with a

shared patient population. The relationship between people and their healthcare environments was the seminal work of White et al. (1961) that introduced the concept of the ecology of medical care (Fryer et al., 2003). This framework used an ecological analysis, which depicted the proportions of people utilizing healthcare services in particular settings, influencing the organization of healthcare, research, and education since its introduction (Fryer et al., 2003).

However, the idea of an ecological examination of healthcare systems or patient populations was not a new concept. Sir William Petty, widely regarded as the father of economics and epidemiology, first suggested comparing outcomes of London and Paris health systems in the seventeenth century (White, 1997). Furthermore, it was J. A. Glover, a British health officer, who documented substantially different rates of tonsillectomies for children during the 1930s based on their location in what were otherwise similar cities or jurisdictions, which was later referred to as the “Glover phenomenon” (White, 1997). The Glover phenomenon is a non-random variation among communities performing elective procedures due to local variations in medical or surgical practices.

The landmark work of White et al. (1961) further asserted that to assess healthcare utilization and the adequacy of medical-care resources, it may be equally important to relate the data to defined populations and health facilities and ask for medical-care decisions. However, ecological analyses are not without criticism because it is impossible to adequately control for confounding effects at the *individual* level, which leads to the “the ecological fallacy.” An ecological fallacy (or ecological inference fallacy) is caused by inferences about individuals that were deduced based on analyses of the group data to which the individual belongs. However, it should also be recognized that some variables can only be measured at

this level since those variables, such as income inequality, are inherent to populations and not individuals, which can lead to a better understanding of population segments (Kaplan, 1996). Thus, ecological analysis is often employed in public health models for health services research in the United States and abroad.

However, until Fryer et al. in 2003, the ecology model had never been applied to understanding disparities nationally in the healthcare system despite its usefulness in conceptualizing healthcare delivery in the United States. The authors assert that advances in U.S. national healthcare surveys permit national inferences based on analyses of population subsets. This allows for an extension of the ecology model, as recommended by its founders, to permit a further examination of the adequacy and appropriateness of existing healthcare resources. The study examined sociodemographic factors on the number of persons per 1,000 in a month who received care in various health settings - a physician's office, outpatient clinic, hospital emergency department, hospital, and their own home (Fryer et al. in 2003).

Ultimately, the authors found that having a usual source of care and health insurance were the primary predictors of the greater likelihood of receiving care in all healthcare settings except emergency departments. This is consistent with prior research that shows the beneficial effect of primary care providing preventative services, resulting in decreased utilization of emergency departments and hospital admissions. While using the ecology model provides a descriptive rather than prescriptive relationship, it does analyze complex relationships known to affect healthcare delivery and population health and provides a framework for policy options and future research design (Fryer et al., 2003).

Another study, by Probst et al., in 2009, examined the association between community health center (CHC) and rural health clinic (RHC) presence and county-

level hospitalization rates for ambulatory care-sensitive conditions in eight states. An ecological analysis was conducted on the relationship between facility presence and county-level hospitalization rates using 2002 hospital discharge data, and the counties were categorized by CHC(s) only, RHC(s) only, both (CHC and RHC), and neither. The study found that a CHC in a county was associated with lower ambulatory care-sensitive hospital admissions rates in working and older adult populations but had no impact on hospitalization rates for the uninsured. During this study, analysts suggested that the rate of FQHC expansion had not kept pace with the rate of increasing uninsured due to insurance erosion. Moreover, the authors posited that a more recent analysis of the same states might produce different results due to the prolific expansion of FQHCs in recent years.

Another ecological study by Rust et al. (2009) examined community health center presence on emergency department (ED) rates in rural counties for the uninsured. The study analyzed data from 117 rural counties in Georgia to determine the relationship between an FQHC in a county and the ED visit rates for the uninsured population. The authors found that counties without an FQHC had a 33% higher rate of ED visits for the uninsured, but no relationship was found for insured patients. Uninsured ED visit rates were also higher for several diagnoses, but particularly for ambulatory care-sensitive conditions. Thus, their findings were consistent with prior studies showing that access to primary care can reduce ED visit rates and that community health centers play an important role in reducing barriers to access, especially for the uninsured, which FQHCs have a unique capacity to serve.

Access to Medical Care Framework

Healthcare access is central to the performance of a health system and has an eminent role in health policy literature (Levesque et al, 2003). Access is synonymous

with the availability of the financial and health system resources in an area (Aday & Andersen, 1974). Healthcare access is defined as the opportunity to receive the appropriate health services for the perceived need of care (Levesque et al, 2003).

Concepts of access in the literature focuses on the characteristics of population (i.e., income, insurance coverage, attitudes towards medical care) or health delivery system i.e., distribution of manpower and facilities). Yet other researchers have suggested that access should be measured by an outcome of the passage of individuals through the healthcare system, such as utilization rates, as a means of external validation of the importance of the individual characteristics or the health system in getting care (Aday & Andersen, 1974).

However, for decades, researchers have failed to account for many related variables – social, economic, structural, and public policy – in which access occurs (Davidson et al, 2004). Thus, a theoretical framework using social indicators and policy viewpoints was developed by Aday & Andersen (1974) for the study of access of medical care. This model of health services usage is an interrelation of variables (i.e., health policy, the health delivery system, and characteristics of the population at risk), but includes external validation of the effect of those variables through utilization of healthcare services and consumer satisfaction (Aday & Andersen, 1974).

The model has been widely accepted as a means to study the impact of a health policy on access to medical care (Maxey et al., 2016). The framework dictates that the policy in question impacts access by enhancing capacity for healthcare delivery via infrastructure, workforces or through social programs like Medicaid (Maxey et al., 2016) It has also been adapted for specific populations at risk, such as the low-income and homeless populations (Davidson et al, 2004; White & Newman, 2015), and to investigate the effect of FQHCs (Maxey et al, 2016, Probst et al, 2009).

A study by Maxey et al (2016) was designed to observe the relationship between health policy and availability of dental services at FQHCs. It posited that FQHCs were the health policy of interest because they represent the vehicle for policies directed at improving health system capacity and access for medically underserved populations. Accordingly, the authors adapted the access to medical care framework to determine the relationship between the policy of interest (FQHCs) and the intended outcome (availability of dental services) (Maxey et al, 2016).

Another study by Probst et al (2009) used the access of medical care framework to conceptualize the use of health services based on external influences (i.e., health services environment, population characteristics, health behavior, and outcomes) to examine the association between the presence of community health centers (CHC) and rural health centers (RHC) and hospitalization rates. It found that hospitalizations rates were reduced in counties with a CHC and counties with both a CHC and RHC, but counties with just an RHC had no effect (Probst et al, 2009). As was found in previous studies, this would suggest that CHC presence is associated with improved access to medical care resulting in better health outcomes for patients and thus are an important part of the healthcare safety net.

Conceptual Model

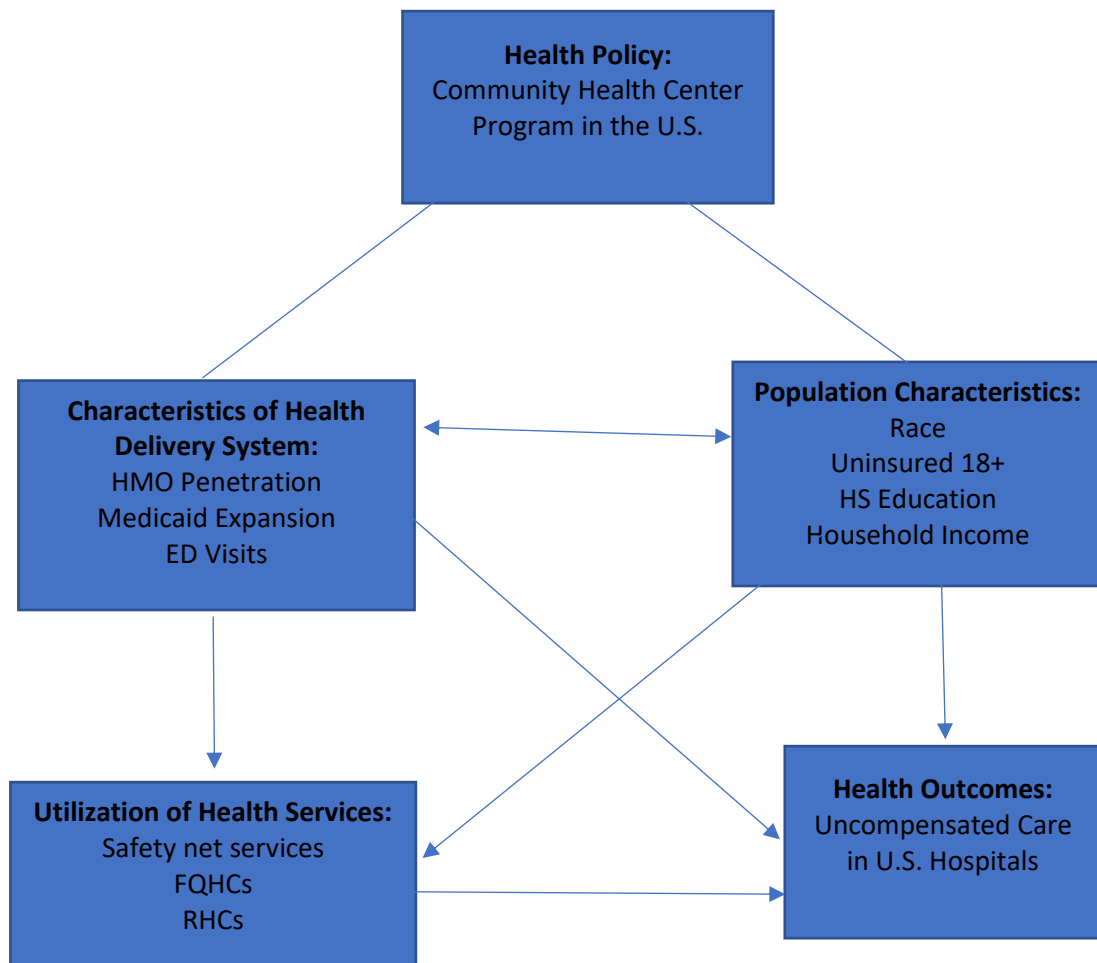


Figure 1. Adapted Aday and Andersen Framework Showing FQHCs as the Federal Policy Vehicle for Access to Medical Care

Research Hypotheses

The literature review above would suggest that uncompensated care may lower in hospitals with a FQHC in the same service area. This study will provide an ecological analysis of the effect of FQHC presence on uncompensated care, which serves as an indicator of access to medical care. While there are many factors that impact the volume of uncompensated care delivery in hospitals, several studies have shown that FQHCs can affect U.S. hospital utilization by decreasing emergency department visits and preventable inpatient admissions (Epstein, 2001; Falik et al, 2001; Probst et al, 2009; Richard et al, 2012). Thus, these reductions of service utilization should lead to a decrease in uncompensated care in hospitals.

In this study, FQHCs are posited as the policy of interest since they often serve as the conduit of health policies aimed at improving access to care for the medically underserved (Maxey et al, 2016). An adapted Aday & Andersen framework (Figure 1) was used to show the relationship between FQHCs, health system characteristics, population characteristics, utilization of health services, and uncompensated care (Aday & Andersen, 1974; Maxey et al, 2016).

Based on prior studies, it is reasonable to assume that FQHCs could potentially impact uncompensated care provided in hospitals within a shared geographical area. By improving access to primary care for low-income and uninsured, FQHCs reduce the demand for more expensive care, such as outpatient specialty care, emergency care, and inpatient hospitalizations, and are subsequently more cost effective (Richards et al, 2012). Ecological studies by Rust et al (2009) and Probst et al (2009) confirmed these findings by showing the impact of community health center presence on the reduction of hospital ED visit rates and ambulatory sensitive hospital admissions.

In developing the questions for this study, it was clear that there was a lack of information on the fiscal impact of community health centers. It was suggested that further research was needed to quantify the economic benefit attributable to community health centers on the reduction of ED visits by the uninsured (Rust et al, 2009). Due to the low-income and uninsured populations that community health centers serve, it seemed logical that the amount of uncompensated care delivered in hospitals would be affected by FQHC presence in a shared geographical area. Thus, this study will determine the economic benefit of community health centers by analyzing the impact of FQHC presence on the level of uncompensated care delivered in U.S. hospitals.

Based on relevant empirical and theoretical literature, the following hypotheses were tested in this study:

1. Hospitals in counties with FQHCs will be associated with lower levels of uncompensated care relative to hospitals in counties without FQHCs.
2. The relationship between FQHC presence and uncompensated care will be stronger due to Medicaid expansion.

CHAPTER 3

METHODOLOGY

Research Methods

This study is a cross-sectional, ecological design that explores the relationship between uncompensated care of hospitals and the presence of a community health center. Using county-level data, the study assesses the effects of FQHC prevalence on uncompensated care for U.S. hospitals. Counties were used because it is the smallest geographical unit available, outside of zip codes and Geocoding coordinates, across the data sources for multivariate analysis. In prior ecological studies, Rust et al., (2009) and Probst et al., (2009), utilized county-level data to examine the impact of community health center presence on hospital rates of ED visits and ambulatory-sensitive hospital admissions. The study data were obtained from the American Hospital Association, Medicare Cost Reports, Uniform Data System, and Area Health Resources Files databases for 2015-2019.

Study Measures

Dependent Variable

Financial information for hospitals was obtained from the Centers for Medicare and Medicaid Services (CMS) Healthcare Cost Report Information System (HCRIS) for 2016. Annual cost reports are submitted by institutional providers to Medicare and are comprised of several worksheets that include information on hospital uncompensated care, utilization statistics, costs, charges, and other financial information. The dependent variable, uncompensated care, was obtained from this data source.

Hospitals' uncompensated care is the sum of bad debt and charity care (Hsieh et al., 2010). Although charity care is theoretically more relevant when examining care for economically disadvantaged patients, researchers have used uncompensated care as a measure instead because hospitals differ considerably in how patients are classified as bad debt or charity care (Hsieh et al, 2010). The hospital's uncompensated care costs were measured by the total of bad debt and charity care. The total sum of uncompensated care provided by hospitals in a county as a whole will then be calculated to provide a measure of the countywide volume of uncompensated care.

Independent Variable

To determine FQHC presence by county, community health center locations were obtained from the Health Center Service Delivery and Look-Alike Sites database from the Health Resource & Service Administration (HRSA) Data Warehouse. For this study, only designated federally qualified community health centers that are permanent service delivery sites were used to determine FQHC penetration, and FQHCs look-alike facilities were excluded. Moreover, health centers designated to serve sub-populations (school health, nursing homes, correctional facilities, and domestic violence) were excluded as they are not intended for general public use. Finally, locations listed as strictly administrative sites were excluded, but combined administrative with service delivery sites were included in this study. This data was obtained from the Uniform Data System (UDS), a standardized reporting system of community health centers and look-alikes' annual performance.

Control Variables

Other variables that could affect uncompensated care were controlled. Based on the access to medical care framework, the control variables represent the health

delivery system, population characteristics, and utilization of health services (Aday & Andersen, 1974). Variables corresponding to the health delivery system and its usage include physician supply, number of hospital beds, number of hospitals with ER departments, ER visits, and HMO penetration (Probst et al., 2009). The American Medical Association Physician Masterfile provided information on physician supply. Hospital characteristics were obtained from the American Hospital Association (AHA). The AHA Annual Survey database is a comprehensive resource for hospital market research and healthcare analysis and contains various organizational and operational data for U.S. hospitals. Also, given the impact of the ACA on uncompensated care, this study controlled for the effect of Medicaid expansion by states.

In prior studies, Rust et al. (2009) and Probst et al. (2009), when examining FQHC presence, used Area Health Resource File (AHRF) for estimates of the uninsured population. AHRF is compiled by HRSA and contains information on more than 50 data sources on healthcare facilities and services, cost and utilization, demographics, and the environment by county and state. AHRF obtains its uninsured data from the U.S. Census, which is conducted every ten years. There are no other publicly available data sources for the uninsured variable updated on an annual basis. Thus, the population characteristics will include controls for the county demand for uncompensated care – the percentage of uninsured, demographic indicators such as the racial and ethnic mix, unemployment rate, and per capita income (Thorpe et al., 2000; Probst et al., 2009). Other county measures include the percentage of the population with less than a high school education, population per square mile, and whether the county is classified as urban or rural, all obtained from AHRF (Probst et al., 2009).

Finally, the utilization of health services was captured by the capacity for safety-net providers in the counties. The examined measures include: (1) the ratio of FQHC visits, (2) the ratio of rural health clinics visits, and (3) the ratio of core safety-net hospital beds in the market to the total population, where increases in these measures indicate more dedicated resources to indigent care (Bazzoli et al., 2006).

Like FQHCs, rural health clinics (RHCs) are located in counties designated as medically underserved but must be in a non-metropolitan area. Although not required to see the uninsured like FQHCs, RHCs receive more patient revenue from the uninsured than FQHCs (Probst et al., 2009). Safety-net hospitals are defined as having a mission to care for individuals regardless of ability to pay and serve a proportionately higher number of uninsured, Medicaid, or otherwise vulnerable patients than their “non-safety net hospital” counterparts (Reiter et al., 2014). Since the amount of care for the uninsured is often correlated with a high volume of Medicaid patients (Hadley & Cunningham, 2004), this study will define safety net hospitals based on hospitals falling in the top quartile of the percentage of patients covered by Medicaid (Reiter et al., 2014).

Table 1 - Summary of Study Variables

Variable Name	Description	Data Format	Data Source
<i>Dependent Variable</i>			
Uncompensated care	Total sum of bad debt and charity care	Continuous	CMS Cost Report
<i>Independent Variable</i>			
FQHCs	Number of FQHCs	Continuous	Uniform Data System
<i>Moderating Variable</i>			
Medicaid expansion	1 = Expanded	Binary	Web source

0 = Non-Expanded

Control Variables

Physician Supply	Number of physicians	Continuous	American Medical Association
Hospital beds	Hospital beds per capita	Continuous	American Hospital Association
ER departments	Hospitals with ER department	Continuous	American Hospital Association
ED visits	ED visits per capita	Continuous	American Hospital Association
HMO penetration	HMO penetration per capita	Continuous	Interstudy

Population Characteristics

Race	Population percentages of racial groups	Continuous	Area Health Resource File
High School Education	Percent with less than HS diploma	Continuous	Area Health Resource File
Unemployed	Percent of population unemployed	Continuous	Area Health Resource File
Uninsured 18+	Percent uninsured, ages 18-64	Continuous	Area Health Resource File
Household income	Median household income	Continuous	Area Health Resource File

Safety Net Services

FQHC visits	Patient visits per capita	Continuous	Area Health Resource File
RHC visits	Patient visits per capita	Continuous	Area Health Resource File
Safety net beds	Hospital beds in safety net hospitals per capita	Continuous	Area Health Resource File

Statistical Analysis

The empirical model examined the effects of FQHC prevalence on uncompensated care provided by hospitals in U.S. counties from 2015-2019. FQHC prevalence was determined by the presence of one or more FQHCs in a county. Since FQHCs are only in specific county types, as required by federal designations, their locations are not random, so differences between the studied counties and counties without FQHCs by controlling for county characteristics (Probst et al., 2009). The total sum of uncompensated care provided by hospitals in each county was summed to provide a measure of the countywide volume of uncompensated care. This county level data was tested to see if FQHC prevalence leads to the reduction of uncompensated care in area hospitals, which would result in a negative coefficient.

Univariate and bivariate analyses were used to describe sample distributions. Based on scatterplot graphs of the main variables of interest, square root transformations were applied to uncompensated care and FQHCs to improve the linear relationship. Moreover, outliers were observed for these variables, but they were determined largely due to geographic distribution. Accordingly, analyses were done further to assess the relationship between uncompensated care and FQHC's presence nationally and in various geographical distributions (i.e., rural, micro area, and metro area) based on population sizes as defined by the USDA Economic Research Service. See Table 2 for the classification of the geographic areas. Furthermore, an additional analysis was done on California and Illinois separately, as those states had FQHCs several times higher than the national mean.

Table 2 - Summary of Geographical Distribution

County Classification	Population Size
Metro area	Urban core of 50,000 or more people
Micro area	Urban core of 2,500-49,9999 people
Rural	Fewer than 2,500 people

Other market conditions were examined for intervening effects on the relationship between FQHCs and uncompensated care. Using an adapted access to medical care framework, covariates were examined based on population characteristics (i.e., race, high school education, unemployment, uninsured 18+, and household income), health delivery system (i.e., Medicaid expansion, HMO penetration, and ED visits), health utilization (FQHCs and RHCs), and health outcomes (uncompensated care). All negative uncompensated care values, representing 2.84% of the sample, were dropped due to assumed reporting errors in the CMS Worksheet S-10 (Crews et al., 2013).

Structural equation modeling (SEM) was utilized to examine the relationship between FQHC prevalence and uncompensated care in hospitals because it permits the examination of multiple relationships simultaneously in a single analysis. Thus, allowing the relationships between multiple variables to be more critically evaluated. Moreover, SEM provided information on model parameters that need re-specification, which allowed for improving the model's fit. As such, some of the originally proposed variables were eliminated due to unsuccessful estimation using SEM as caused by multicollinearity. See the correlation matrix in Appendix A. Finally, an additional model was fitted with Medicaid expansion as a moderator to examine further the primary relationship and covariates that may be impacted by its effect.

The statistical analyses were conducted using Stata version 17.0, and p-values less than or equal to 0.05 was considered statistically significant. A priori power

analysis for SEMs was conducted based on the number of variables in the model, with a small-anticipated effect size ($d=.10$) and an alpha of .05. The results showed that a minimum sample size of 1,258 participants was required to achieve a power of .80. The maximum likelihood estimation method was used for model specification and model estimation.

CHAPTER 4

RESULTS

This chapter presents the analysis results, which examine the hypotheses of this study. As described in Chapter 3, three sets of analyses were conducted. This included a primary analysis of the effect of FQHC prevalence on uncompensated care in U.S. hospitals. In addition, two subgroup analyses were conducted to examine this relationship in various geographic areas and analyze the effect of Medicaid expansion as a moderator in the model.

Descriptive Statistics

Application of the variable selection for this study resulted in 25,147 observations for the years, 2015 to 2019. Of which, 79.32% of the counties had one or more FQHCs and 20.68% of counties did not have a single FQHC. Moreover, 79.16% of counties reported having uncompensated care, 18.00% reported zero uncompensated care, and 2.84% reported negative uncompensated care.

Furthermore, there was an average of 13.63 FQHCs and \$7,177,138 of uncompensated care costs per county nationally. Table 3 summarizes the mean, standard deviations, minimum and maximum values for all examined variables in this study.

Table 3 - Descriptive Statistics for National Data

Variable	Mean	Std. dev.	Min	Max
Uncompensated care	7177138	4.95e+07	0	6.70e+09
FQHCs	13.63419	33.02158	0	252
Medicaid expansion	.5571721	.4967306	.16	1
ED visits per 10,000	275717.2	517143.1	0	3199835
HMO penetration	30.53862	15.48866	0	88.73
Household income	56563.73	16406.77	0	151806
Population estimate	687155.3	1509397	0	1.02e+07
Percent uninsured, age 18-64	10.3802	5.019131	0	35
Percent uninsured, age 17 or less	4.997972	3.086667	0	30
Percent of population that is unemployed	5.436634	2.208933	0	31.01154
Percent with less than high school education	12.29803	5.44895	1.9	46.7
Percent of population that is				
White	67.1898	22.47782	0	98.05807
Black	11.30562	13.63748	0	85.87005
Asian	3.604548	5.074321	0	43.00673
Latino	12.16415	14.72165	0	95.44462
Native American	1.794927	5.694492	0	87.97814

Regression Analysis

Based on relevant empirical and theoretical literature, structural equation modeling (SEM) was used to test the following hypotheses:

1. Hospitals in counties with FQHCs will be associated with lower levels of uncompensated care relative to hospitals in counties without FQHCs.
2. The relationship between FQHC presence and uncompensated care will be stronger due to Medicaid expansion.

Analysis for Hypothesis 1 - FQHCs Prevalence to Uncompensated Care

Due to the high skewness of cost and the distribution of FQHCs nationally, square root transformations were done on the FQHCs and uncompensated care cost variables. The results of the SEM analysis are shown in Table 4. All of the covariates were statistically significant except for Latino and Asian populations. The hypothesis stated that hospitals with FQHCs would be associated with lower levels of uncompensated care, which was not supported. Based on the results, counties with FQHCs ($\beta = .1251405$, $SE = .013989$, $P = .0977225$) costs are higher as indicated by a positive coefficient. Of the covariates, only RHCs, ED visits, White and Native American demonstrate an inverse relationship to uncompensated care. Thus, effectively lowering costs based on national data.

Table 4 - Analysis of Hypotheses 1 – FQHCs Prevalence to Uncompensated Care

Effect sqrtUncompensated care	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
sqrtFQHC***	.1251405	.013989	.0977225	.1525585	0.000
RHC*	-.0134881	.0066425	-.0265071	-.000469	0.042
ED visits***	-.0631156	.0129676	-.0885317	-.0376995	0.000
HMO penetration***	.0613051	.0069293	.047724	.0748863	0.000

High school***	.097112	.0101864	-.117077	-.0771469	0.000
Unemployment rate**	.0303252	.0091745	.0123436	.0483069	0.001
Uninsured, age 18-64***	.1607627	.0090208	.1430822	.1784431	0.000
Household income***	.0980552	.0103502	.0777691	.1183413	0.000
White**	-.05397	.0201915	-.0935445	-.0143954	0.008
Black***	.0929231	.0128477	.067742	.1181042	0.000
Latino	.0000662	.0168243	-.0329088	.0330411	0.997
Asian	.0101156	.0113338	-.0120983	.0323295	0.372
Native American***	-.0745048	.008144	-.0904668	-.0585428	0.000

*p<0.05, **p<0.01, ***p<0.001

Sub-Analysis of Hypotheses 1 – Geographic Distributions

As previously noted, there were outliers found in the national dataset. Based on further examination of the data, it was found that the skewness was along geographical distributions (rural, micro metro, and metropolitan). Moreover, a separate analysis was done for California and Illinois because those combined states had a mean of 65.24 FQHCs, which is over five times the mean nationally. The results for the geographical distributions can be found in Tables 5-8.

It was observed that FQHCs were statistically significant in all of the geographical distributions, except rural, and were associated with an increase in uncompensated care as observed with the national data. However, there were important differences to note based on geography. In rural areas, there were no statistically significant factors to reduce uncompensated care. Only ED visits and HMO penetration were statistically significant. Again, this was the only geographic

distribution that FQHCs were not statistically significant ($\beta = .0235687$, $SE = .0405401$, $P = 0.561$). However, this would be expected due to their low prevalence in rural areas, comparatively to RHCs, which were surprisingly also not statistically significant ($\beta = .0292867$, $SE = .0376796$, $P = 0.437$).

Table 5 - Rural Analysis

Effect	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
sqrtUncompensated care					
sqrtFQHC	.0235687	.0405401	.0558884	.1030259	0.561
RHC	.0292867	.0376796	-.044564	.1031373	0.437
ED visits*	.5002488	.2156866	.0775108	.9229868	0.020
HMO penetration*	.0672576	.0318346	.0048629	.1296523	0.035
High school	-.0438702	.0589018	.1593156	.0715752	0.456
Unemployment rate	.0080393	.0549377	.0996366	.1157153	0.884
Uninsured, age 18-64***	.1739575	.0469973	.0818445	.2660705	0.000
Household income	.0009153	.0526607	.1022977	.1041283	0.986
White	-.8165914	.7002851	2.189125	.5559423	0.244
Black	-.1776208	.5457257	1.247223	.8919819	0.745
Latino	-.4166989	.5402616	1.475592	.6421944	0.441
Asian	.0711523	.0536527	.0340051	.1763096	0.185
Native American	-.0157761	.0598634	.1331063	.1015541	0.792

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

For micro metro areas, there were several variables, including FQHCs, that were statistically significant. However, only one that reduced the cost of

uncompensated care, which was the population with less than a high school education. In metropolitan areas, several covariates were statistically significant, including FQHCs ($\beta = .0318235$, $SE = .011973$, $P = .0008$), but only ED visits, White and Native American populations had negative coefficients. By examining different geographic distributions, it was noted that there was a steady increase of FQHCs prevalence effect on uncompensated care as the mean of FQHCs increased in counties.

Table 6 - Micro Metro Analysis

Effect	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
sqrtUncompensated care					
sqrtFQHC**	.0318235	.011973	.0083568	.0552902	0.008
RHC	.0015765	.0110319	-.0200455	.0231985	0.886
ED visits***	.2448244	.0109047	.2234516	.2661973	0.000
HMO penetration	.0163608	.0113507	-.0058861	.0386077	0.149
High school*	-.0416028	.0176743	-.0762438	-.0069618	0.019
Unemployment rate**	.041426	.0148187	.012382	.0704701	0.005
Uninsured, age 18-64***	.1820723	.0156974	.1513061	.2128386	0.000
Household income***	.0647755	.0148803	.0356106	.0939405	0.000
White	.0037411	.1921177	-.3728028	.3802849	0.984
Black	.0426503	.1478609	-.2471517	.3324522	0.773
Latino	.0210506	.1249977	-.2239404	.2660416	0.866
Asian**	.0433336	.0166357	.0107283	.075939	0.009
Native American	-.0964851	.0770182	-.247438	.0544677	0.210

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7 - Metropolitan Analysis

Effect sqrtUncompensated care	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
sqrtFQHC***	.0632665	.0172435	.0294699	.0970631	0.000
RHC	-.0133822	.0084991	.0300402	.0032758	0.115
ED visits*	-.0399842	.0165956	.0725109	-.0074575	0.016
HMO penetration*	.0211523	.0082369	.0050083	.0372963	0.010
High school	-.0240908	.0147469	.0529942	.0048127	0.102
Unemployment rate	-.0067184	.0119068	.0300554	.0166185	0.573
Uninsured, age 18-64***	.1560211	.0114983	.1334848	.1785573	0.000
Household income***	.0646846	.0128666	.0394665	.0899027	0.000
White*	-.1364377	.0690709	.2718141	-.0010612	0.048
Black	.0340947	.0428177	.0498264	.1180158	0.426
Latino	-.1039877	.0534302	.2087089	.0007334	0.052
Asian	-.0211541	.0279714	-.075977	.0336687	0.449
Native American***	-.0775473	.0171973	.1112534	-.0438411	0.000

*p<0.05, **p<0.01, ***p<0.001

The analysis of California and Illinois showed several population and health delivery system differences versus the nation. Aside from having over five times the mean of FQHCs, these states had over three times the number of ED visits, but yet lower uncompensated care costs. This is likely due to higher median incomes, rates of insurance and HMO penetration than seen nationally. Despite these advantages, the presence of FQHCs in California and Illinois (β =.2454458, SE=.0782891,

P=.0.002) had nearly double the effect on uncompensated care than the national average. Thus, further suggesting that FQHCs prevalence has a positive relationship with uncompensated care in U.S. hospitals. In these states, only HMO penetration and ED visits were associated with a reduction of uncompensated care cost. Furthermore, the effect was three times greater ($\beta = -.2036317$ versus $\beta = -.0631156$), which was comparable to the volume increase (959,176.8 versus 275,717.2) of ED visits over the national mean.

Table 8 - Descriptive Statistics for California and Illinois

Variable	Mean	Std. dev.	Min	Max
Uncompensated care	6255547	2.35e+07	-9098662	5.47e+08
FQHCs	65.24418	80.54756	0	252
ED visits per 10,000	959176.8	1142426	974	3199835
HMO penetration	33.92275	15.74478	1.23	60.88
Median household income	66622.79	17225.91	34982	135234
Population estimate	2831708	3509926	3821	1.02e+07
Percent uninsured, age 18-64	7.867068	2.138721	4	13
Percent uninsured, age 17 or less	2.774699	.6691882	1	6
Percent of population that is unemployed	6.140543	1.607512	2.584721	14.17755
Percent with less than high school education	14.47434	5.490644	4.1	30.9
Percent of population that is				
White	51.34728	22.72694	10.0218	97.22823

Black	7.673293	6.821209	.3403707	30.28511
Asian	9.944264	8.551211	.2804878	39.01959
Latino	25.53381	16.39168	.8840186	80.17162
Native American	1.38609	1.230278	.1387312	13.48752

Table 9 - California and Illinois Analysis

Effect	Estimate	SE	95% CI		p
	Standardized Coefficient (β)		LL	UL	
sqrtUncompensated care					
sqrtFQHC**	.2454458	.0782891	.092002	.3988896	0.002
RHC	-.0150915	.0283824	-.07072	.040537	0.595
ED visits*	-.2036317	.0787581	-.3579947	-.0492688	0.010
HMO penetration*	-.0688923	.0322092	-.1320211	-.0057634	0.032
High school	-.1072783	.0555771	-.2162074	.0016508	0.054
Unemployment rate	.0551768	.0332146	-.0099227	.1202762	0.097
Uninsured, age 18-64	.0038639	.0445521	-.0834565	.0911844	0.931
Household income*	.1140186	.0464853	.0229091	.2051281	0.014
White	.7301414	.4968015	-.2435716	1.703854	0.142
Black*	.3959897	.1547504	.0926845	.6992949	0.011
Latino	.6466329	.3815189	-.1011304	1.394396	0.090
Asian	.3365388	.2129025	-.0807424	.7538201	0.114
Native American	.0219296	.0491221	-.0743479	.1182072	0.655

*p<0.05, **p<0.01, ***p<0.001

Analysis for Hypothesis 2 – Effect of Medicaid Expansion

The second analysis was the effect of Medicaid expansion on FQHCs presence and uncompensated care cost. Based on the literature review and theoretical framework, the hypothesis was that the relationship between FQHCs prevalence and uncompensated care would be strengthened by Medicaid expansion. It was statistically significant and marginally supported ($\beta = .1276669$, $SE = .0143667$, $P = 0.000$) when compared the previous model without Medicaid expansion ($\beta = .1251405$, $SE = .013989$, $P = 0.000$) nationally. However, Medicaid expansion had a greater effect on other covariates, uninsured adults, HMO penetration, and ED visits. The full results can be found in Table 10.

Table 10 - Regression Analysis of Hypothesis 2 – Effect of Medicaid Expansion

Effect sqrtUncompensated care	Estimate Standardized Coefficient (β)	SE	95% CI		<i>p</i>
			<i>LL</i>	<i>UL</i>	
sqrtFQHC***	.1276669	.0143667	.0995086	.1558252	0.000
RHC**	-.0188074	.00668	-.0319	-.0057148	0.005
Medicaid expansion***	-.085171	.0090858	-.1029788	-.0673632	0.000
ED visits***	-.0537803	.0134305	-.0801037	-.0274569	0.000
HMO penetration***	.0592831	.0067775	.0459993	.0725668	0.000
High school***	-.077013	.0103744	-.0973465	-.0566794	0.000
Unemployment rate***	.0432038	.0087215	.02611	.0602976	0.000
Uninsured, age 18- 64***	.0864433	.0118013	.0633132	.1095734	0.000
Household income***	.0947204	.0100574	.0750082	.1144326	0.000
White**	-.218088	.0658324	-.3471172	-.0890589	0.001

Black	-.0280636	.0417539	-.1098997	.0537725	0.502
Latino*	-.1023474	.0464808	-.1934481	-.0112467	0.028
Asian	-.0413616	.0229447	-.0863323	.0036092	0.071
Native American***	-.1107834	.0185879	-.1472149	-.0743518	0.000

*p<0.05, **p<0.01, ***p<0.001

Based on the above results, Medicaid expansion seemed to have moderating effect on the some of the covariates in this study. As such, a model was run with Medicaid expansion as a moderator for the following variables, FQHCs, uninsured adults, HMO penetration, and ED visits. All of the relationships were statistically significant and Medicaid expansion even changed the direction of the relationship between uncompensated care with ED visits and uninsured adults, which supports its effect as a moderator for this study. The results of the Medicaid expansion as a moderator can be found in Table 11.

Table 11 - Medicaid as a Moderator Analysis

Effect sqrtUncompensated care	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
<i>with Moderator – Medicaid expansion</i>					
sqrtFQHC***	.2127784	.0059864	.2010453	.2245114	0.000
ED visits***	.1320679	.0062046	.1199071	.1442287	0.000
HMO penetration***	.097494	.0062672	.0852105	.1097776	0.000
Uninsured, age 18- 64***	-.6300112	.0034245	-.6367232	-.6232993	0.000
<i>without Moderator - Medicaid expansion</i>					

sqrtFQHC***	.176313	.0126624	.1514951	.2011309	0.000
ED visits***	-.0559289	.0128399	-.0810946	-.0307633	0.000
HMO penetration***	.085929	.0066996	.072798	.09906	0.000
Uninsured, age 18- 64***	.0940492	.0062441	.0818109	.1062874	0.000

*p<0.05, **p<0.01, ***p<0.001

Lastly, the final analysis included a year-by-year comparison, 2015-2019, of the effect of FQHCs presence on uncompensated care costs. The relationship with FQHCs to uncompensated care remained statistically significant and positive through all the years of this study. The effect of Medicaid expansion was also significant and steadily reduced uncompensated care costs until a slight dip of its effect in 2019. The results of this analysis can be found in Table 12.

Table 12 - By Year Analysis, 2015-2019

Effect	Estimate Standardized Coefficient (β)	SE	95% CI		p
			LL	UL	
sqrtUncompensated care					
2015					
sqrtFQHC***	.1186428	.0313982	.0571034	.1801821	0.000
Medicaid expansion*	-.0423829	.0212688	-.084069	-.0006968	0.046
2016					
sqrtFQHC***	.1340722	.0339313	.067568	.2005764	0.000
Medicaid expansion**	-.0667153	.0203595	-.1066192	-.0268114	0.001
2017					
sqrtFQHC***	.1217263	.0321704	.0586736	.1847791	0.000

Medicaid expansion***	-.0879703	.0205948	-.1283355	-.0476052	0.000
<i>2018</i>					
sqrtFQHC***	.1481625	.0327361	.0840008	.2123241	0.000
Medicaid expansion***	-.1065466	.0199262	-.1456011	-.067492	0.000
<i>2019</i>					
sqrtFQHC**	.1136858	.0330608	.0488878	.1784839	0.001
Medicaid expansion***	-.0987828	.0202	-.1383742	-.0591915	0.000

*p<0.05, **p<0.01, ***p<0.001

CHAPTER 5

DISCUSSION

Uncompensated care is a complex issue, but it is typically affected by policy and market factors such as legislative changes and an increase in the number of uninsured. Although hospitals continue to serve as critical safety-net providers, they are often not the most appropriate facility to serve the indigent population. Instead, federally qualified community health centers (FQHCs) were designed to provide care for the poor and uninsured. Yet there is minimal literature on the impact of FQHC penetration on hospitals, particularly regarding economic benefits. This study aimed to examine FQHC's presence on uncompensated care in U.S. hospitals. The theoretical framework of access to medical care, as well as the ecology of medical care, were utilized to create the following research questions:

1. Does access to medical care at FQHCs affect uncompensated care provided in U.S. hospitals?
2. Does Medicaid expansion affect the relationship between FQHC presence and uncompensated care in U.S. hospitals?

Review and Assessment of Findings

Based on this analysis, FQHCs presence does affect uncompensated care in U.S. hospitals, but it does not support the hypothesis that hospitals in counties with FQHCs will experience lower uncompensated care. Our regression analysis consistently found a statistically significant positive relationship between FQHC prevalence and uncompensated care in U.S. hospitals. Although surprising, this study consistently showed that FQHCs' presence increased uncompensated care every year

examined by this study. Furthermore, in all types of geographic distributions (rural, micro metro, or metropolitan), the presence of FQHCs in counties positively affected uncompensated care. The strength of that relationship substantially increased in California and Illinois, which have the highest concentration of FQHCs.

However, it may not just be the proximity of FQHCs to area hospitals that is driving this inverse relationship with uncompensated care. It is a common practice for these facilities to partner in treating indigent populations. This reciprocal relationship allows FQHCs to provide the necessary primary care services, but they typically refer to partner hospitals for specialty and inpatient care for their patients with more complex cases, which could result in higher uncompensated care costs for area hospitals due to higher resource consumption.

Moreover, many environmental and policy factors impact uncompensated care, which could supersede the effects of FQHC prevalence. Due to the Patient Protection and Affordable Care Act (ACA) implementation, uncompensated care in U.S. hospitals started to decline in 2014 but began to rise again in 2016. Likewise, Medicaid expansion was implemented in most states in 2014, just before this study's timeframe in 2015-2019. As such, we also examined the effect of Medicaid expansion.

As stated in the second hypothesis, the findings showed that the relationship between FQHCs presence and uncompensated care was stronger in states that had expanded. This effect was further amplified in California and Illinois, the states with Medicaid expansion and the highest mean of FQHCs. The directionality remained the same. Thus, FQHCs presence seems to increase uncompensated care in hospitals in both expanded and non-expanded states.

To further understand this relationship, the model was fitted with Medicaid expansion as a moderator to examine the variables that its effect would most impact. Again, Medicaid expansion amplified the positive relationship between uncompensated care and FQHCs, as well as HMO penetration. The strengthened impact of Medicaid expansion was not a surprise because the ACA encouraged the interdependence of health facilities, including hospitals and FQHCs, via the creation of accountable care organizations (ACOs) networks (Lewis et al.). However, Medicaid expansion changed the directionality of the relationship between uncompensated care with ED visits and uninsured, which supports its effect as a moderator in this study. Moreover, all the associations were statistically significant, with Medicaid expansion as a moderator.

Although the hypotheses' directionality regarding the primary relationship of this study was not supported, it may be important to note that the theoretical frameworks on which this study was built are more descriptive than prescriptive. Therefore, they are employed to analyze complex relationships that are known to affect healthcare delivery and population health (Fryer et al., 2003). As such, the ecological effect of medical care was confirmed since the relationship between FQHCs presence and uncompensated care in U.S. hospitals was statistically significant in every geographical distribution examined in this study, except rural counties, where FQHCs were less prevalent.

Study Limitations

This study had several limitations due to the availability and reliability of data. First, some of the proposed data was largely missing, which affected the design of this study. Initially, FQHC patient visits data was proposed as a covariate to examine whether FQHC service capacity affected uncompensated care. Unfortunately, this

data is largely missing from the UDS dataset, which did not allow this examination. Although we cannot speculate on the directionality, having the capacity data (as measured by patient visits at the county level) may have provided further insight into the effect of FQHCs presence on uncompensated care.

Second, the data utilized for this analysis was primarily collected for administrative purposes. As such, there is the potential for reporting variability and inaccuracy, particularly for the dependent variable. Although charity care cost is theoretically more relevant, researchers have used uncompensated care as a measure because hospitals differ considerably on how patients are classified as bad debt versus charity care (Hsieh et al., 2010). Moreover, there were several data quality issues identified with reporting in the CMS Worksheet S-10, where uncompensated care data is captured, which could make the use of this data imprecise for research purposes (Crews et al., 2013).

Third, the method of this study also presented a limitation because it examined the relationship between FQHC's presence and uncompensated care at the county level. Ideally, examining the relationship between these facilities directly, not as a total of these variables at the county level, may have provided a more robust analysis. However, there is limited reporting data available in secondary datasets regarding the connections between FQHCs and hospitals.

Lastly, SEMs are highly sensitive to normality and multicollinearity, so square root transformations were applied to the main variables of interest. Uncompensated care is a cost-based outcome, which is usually highly skewed. Moreover, there were observed outliers in the distribution of FQHCs nationally. Despite these transformations, it is possible that normality was not perfectly achieved. Due to multicollinearity, some of the proposed covariates were dropped, which may have

been equally or better suited as measures for this model. Furthermore, due to the complexity of uncompensated care, there could be other factors not considered by this study that may provide more insight.

Recommendations for Future Research

Due to findings that FQHCs are positively correlated with uncompensated care in U.S. hospitals, future research is needed to understand the relationship between these healthcare facilities better. Although several studies have shown a reduction in emergency department visits and preventable hospitalizations (Epstein, 2001; Falik et al., 2001; Probst et al., 2009; Richard et al., 2012), it was interesting to discover that FQHC presence did not reduce hospitals' uncompensated care accordingly. Unlike this study, the incongruence may be due to prior studies examining FQHCs' effect within a limited geographical area, such as a particular city or state(s).

Although this study examined the relationship between FQHCs' presence and hospitals' uncompensated care nationally, this resulted in county-level measurements. A more robust analysis would provide a direct correlation between these healthcare facilities. This could be done using zip codes with geographic information system (GIS) mapping to narrow the proximity between FQHCs and hospitals further. Likewise, utilizing data from CMS' Medicare Shared Savings Program for Accountable Care Organizations (ACOs), which lists all the ACO participants by ACO name and organizational information, could provide a direct connection between FQHCs and hospitals for future analysis of uncompensated care. Finally, examining healthcare utilization patterns, such as patient visits and referrals, between FQHCs and hospitals may help to further inform researchers about the findings of this study.

Conclusion

This study aimed to examine the relationship between FQHCs prevalence and uncompensated care in U.S. hospitals. Despite prior studies examining other dependent measures such as ED visits and hospitalizations (Epstein, 2001; Falik et al., 2001; Probst et al., 2009; Richard et al., 2012), there was a gap in the literature regarding the economic effects of this relationship. Findings in this analysis showed that FQHC presence in counties increased the odds of uncompensated care in area hospitals, further magnified by the effect of Medicaid expansion. Moreover, Medicaid expansion was an effective moderator across selected covariates as this relationship is multifaceted. Thus, understanding what truly drives uncompensated care in U.S. hospitals remains a complex issue that goes beyond a single study's reach. Future studies may consider different market and policy factors as exploratory variables to further expand the scope of knowledge in this field.

REFERENCES

- Aday, L. A., & Andersen, R. (1974). A framework for the study of access to medical care. *Health Services Research*, 9(3), 208–220.
<https://www.ncbi.nlm.nih.gov/pubmed/4436074>
- Aldrich, H. E., & Pfeffer, J. (1976). Environments of organizations. *Annual Review of Sociology*, 2(1), 79–105. doi:10.1146/annurev.so.02.080176.000455
- American Hospital Association Annual Survey (2017). <http://www.khanet.org/Communications/CurrentReportArticles/2017-American-Hospital-Association-Annual-Survey>
- American Hospital Association, Uncompensated care cost fact sheet. (2017).
<https://www.aha.org/system/files/2018-01/2017-uncompensated-care-factsheet.pdf>
- Asplin, B. R. (2001). Tying a knot in the unraveling health care safety net. *Academic Emergency Medicine: Official Journal of the Society for Academic Emergency Medicine*, 8(11), 1075–1079. doi:10.1111/j.1553-2712.2001.tb01118.x
- Auter, Z. (2018). US Uninsured Rate Steady at 12.2% in Fourth Quarter of 2017. Gallup. 2018; published online January 16. <https://news.gallup.com/poll/225383/uninsured-rate-steady-fourth-quarter-2017.aspx>
- Banks, D. A., Paterson, M., & Wendel, J. (1997). Uncompensated hospital care: charitable mission or profitable business decision? *Health Economics*, 6(2), 133–143. doi:10.1002/(sici)1099-1050(199703)6:23.0.co;2-x
- Baum, J. A. C., & Shipilov, A. V. (2006). Ecological approaches to organizations. *The SAGE Handbook of Organization Studies* (55–110). doi:10.4135/9781848608030.n3
- Bazzoli, G. J., Lindrooth, R. C., Kang, R., & Hasnain-Wynia, R. (2006). The influence of health policy and market factors on the hospital safety net. *Health Services Research*, 41(4 Pt 1), 1159–1180. doi:10.1111/j.1475-6773.2006.00528.x

- Blavin, F. (2016). Association between the 2014 Medicaid expansion and US hospital finances. *JAMA: The Journal of the American Medical Association*, 316(14), 1475. doi:10.1001/jama.2016.14765
- Camilleri, S. (2018). The ACA Medicaid expansion, disproportionate share hospitals, and uncompensated care. *Health Services Research*, 53(3), 1562–1580. doi:10.1111/1475-6773.12702
- Crews, C. Randolph, R., Holmes, Pink, G., Reiter K. L. (2013). Provision of uncompensated care by rural hospitals: a preliminary look at Medicare cost report worksheet S-10. *North Carolina Rural Health Research*.
- Center on Budget and Policy Priorities. Health insurance coverage. Uncompensated care costs fell in nearly every state as ACA's major coverage provisions took effect (2018). Washington, D.C.
- Center for Disease Control and Prevention. National Center for Health Statistics. (2017) <https://www.cdc.gov/nchs/fastats/health-insurance.htm>
- Chang, C. H., Bynum, J., & Lurie, J. D. (2019). Geographic Expansion of Federally Qualified Health Centers 2007-2014. *The Journal of Rural Health: Official Journal of the American Rural Health Association and the National Rural Health Care Association*. 35, 385–394. doi:10.1111/jrh.12330
- Coughlin, T. A., Holahan, J., Caswell, K., & McGrath, M. (2014). Uncompensated care for the uninsured in 2013: A detailed examination. KFF website: <https://www.kff.org/uninsured/report/uncompensated-care-for-the-uninsured-in-2013-a-detailed-examination/view/footnotes/>
- Cunningham, P. J., Bazzoli, G. J., & Katz, A. (2008). [Review of Caught in the competitive crossfire: safety-net providers balance margin and mission in a profit-driven health care market]. *Health affairs (Project Hope)*, 27(5), w374-82. doi:10.1377/hlthaff.27.5.w374
- Davidson, P. L., Andersen, R. M., Wyn, R., & Brown, E. R. (2004). A framework for evaluating safety-net and other community-level factors on access for low-income populations. *Inquiry: A Journal of Medical Care Organization, Provision and Financing*, 41(1), 21–38. doi:10.5034/inquiryjrnl_41.1.21
- Dranove, D., Garthwaite, C., & Ody, C. (2016). Uncompensated care decreased at hospitals in Medicaid expansion states but not at hospitals in nonexpansion states. *Health Affairs (Project Hope)*, 35(8), 1471–1479. doi:10.1377/hlthaff.2015.1344
- Falik, M., Needleman, J., Wells, B. L., & Korb, J. (2001). Ambulatory care sensitive hospitalizations and emergency visits: *Medical Care*, 39(6), 551–561. doi:10.1097/00005650-200106000-00004

- Ferrier, G. D., Rosko, M. D., & Valdmanis, V. G. (2006). Analysis of uncompensated hospital care using a DEA model of output congestion. *Health Care Management Science*, 9(2), 181–188. doi:10.1007/s10729-006-7665-8
- Forrest, C. B., & Whelan, E. M. (2000). Primary care safety-net delivery sites in the United States: A comparison of community health centers, hospital outpatient departments, and physicians' offices. *JAMA: The Journal of the American Medical Association*, 284(16), 2077–2083. doi:10.1001/jama.284.16.2077
- Fryer, G. E., Jr, Green, L. A., Dovey, S. M., Yawn, B. P., Phillips, R. L., & Lanier, D. (2003). Variation in the ecology of medical care. *Annals of Family Medicine*, 1(2), 81–89. doi:10.1370/afm.52
- Hadley, J., & Cunningham, P. (2004). Availability of safety net providers and access to care of uninsured persons. *Health Services Research*, 39(5), 1527–1546. doi:10.1111/j.1475-6773.2004.00302.x
- Hannan, M. T., & Freeman, J. (1977). The population ecology of organizations. *American journal of sociology*, 82(5), 929–964. doi:10.1086/226424
- Hesketh, M. &, & Dwiel, K. (2017). Background note on federally qualified health centers (FQHC). *Harvard Medical School Center for Primary Care*.
- Hsieh, H.-M., Clement, D. G., & Bazzoli, G. J. (2010). Impacts of market and organizational characteristics on hospital efficiency and uncompensated care. *Health Care Management Review*, 35(1), 77–87. doi:10.1097/HMR.0b013e3181c09956
- Jedele, J. (2016). Federally qualified health centers in a changing health care environment: are they prepared for the challenge? (Doctoral dissertation) https://deepblue.lib.umich.edu/bitstream/handle/2027.42/135787/jwillem_1.pdf
- Kaiser Family Foundation. Status of state action on the Medicaid expansion decision. (2018). KFF website: <https://www.kff.org/health-reform/state-indicator/state-activity-around-expanding-medicaid-under-the-affordable-care-act/>
- Kaplan, G. A., Pamuk, E. R., Lynch, J. W., Cohen, R. D., & Balfour, J. L. (1996). Inequality in income and mortality in the United States: analysis of mortality and potential pathways. *BMJ (Clinical Research Ed.)*, 312(7037), 999–1003. doi:10.1136/bmj.312.7037.999
- Ku, L., Richard, P., Dor, A., Tan, E., Shin, P., & Rosenbaum, S. (2009). Using primary care to bend the curve: Estimating the impact of a health center expansion on health care costs. (*Geiger Gibson/RCHN Community Health Foundation Research Collaborative policy*).
- LaCalle, E., & Rabin, E. (2010). Frequent users of emergency departments: the myths, the data, and the policy implications. *Annals of Emergency Medicine*, 56(1), 42–48. doi:10.1016/j.annemergmed.2010.01.032

- Laiteerapong, N., Kirby, J., Gao, Y., Yu, T.-C., Sharma, R., Nocon, R., ... Huang, E. S. (2014). Health care utilization and receipt of preventive care for patients seen at federally funded health centers compared to other sites of primary care. *Health Services Research*, 49(5), 1498–1518. doi:10.1111/1475-6773.12178
- Levesque, J.-F., Harris, M. F., & Russell, G. (2013). Patient-centred access to health care: conceptualising access at the interface of health systems and populations. *International Journal for Equity in Health*, 12(1), 18. doi:10.1186/1475-9276-12-18
- Lewis, V. A., Colla, C. H., Schoenherr, K. E., Shortell, S. M., & Fisher, E. S. (2014). Innovation in the safety net: integrating community health centers through accountable care. *Journal of general internal medicine*, 29(11), 1484–1490. <https://doi.org/10.1007/s11606-014-2911-0>
- MACPAC Annual Analysis of Disproportionate Share Hospital Allotments to States. (2018). *MACPAC, Medicaid and CHIP Payment and Access Commission*. U.S.; Washington, DC.
- Mann, J., Melnick, G., Bamezai, A., & Zwanziger, J. (1995). Uncompensated care: hospitals' responses to fiscal pressures. *Health Affairs (Project Hope)*, 14(1), 263–270. doi:10.1377/hlthaff.14.1.263
- Mann, J. M., Melnick, G. A., Bamezai, A., & Zwanziger, J. (1997). A profile of uncompensated hospital care, 1983-1995. *Health Affairs (Project Hope)*, 16(4), 223–232. doi:10.1377/hlthaff.16.4.223
- May 2018 Medicaid & CHIP enrollment data highlights. <https://www.medicaid.gov/medicaid/program-information/medicaid-and-chip-enrollment-data/report-highlights/index.html>
- Maxey, H. L., Norwood, C. W., & Liu, Z. (2016). State policy environment and the dental safety net: a case study of professional practice environments' effect on dental service availability in Federally Qualified Health Centers. *Journal of Public Health Dentistry*, 76(4), 295–302. doi:10.1111/jphd.12155
- Gonzalez Morganti, K., Bauhoff, S., Blanchard, J. C., Abir, M., Iyer, N., Smith, A. C., Kellermann, A. L. (2013). *The evolving role of emergency departments in the United States*. https://www.rand.org/pubs/research_reports/RR280.html
- NACHC About health centers. (2015). NACHC website: <http://www.nachc.org/about-our-health-centers/>
- Nikpay, S., Buchmueller, T., & Levy, H. (2015). Early Medicaid expansion in Connecticut stemmed the growth in hospital uncompensated care. *Health Affairs (Project Hope)*, 34(7), 1170–1179. doi:10.1377/hlthaff.2015.0107
- Probst, J. C., Laditka, J. N., & Laditka, S. B. (2009). Association between community health center and rural health clinic presence and county-level hospitalization

- rates for ambulatory care sensitive conditions: an analysis across eight US states. *BMC Health Services Research*, 9(1), 134. doi:10.1186/1472-6963-9-134
- Reiter, K. L., Jiang, H. J., & Wang, J. (2014). Facing the recession: how did safety-net hospitals fare financially compared with their peers? *Health Services Research*, 49(6), 1747–1766. doi:10.1111/1475-6773.12230
- Richard, P., Ku, L., Dor, A., Tan, E., Shin, P., & Rosenbaum, S. (2012). Cost savings associated with the use of community health centers. *The Journal of Ambulatory Care Management*, 35(1), 50–59. doi:10.1097/JAC.0b013e31823d27b6
- Rothkopf, J., Brookler, K., Wadhwa, S., & Sajovetz, M. (2011). Medicaid patients seen at federally qualified health centers use hospital services less than those seen by private providers. *Health Affairs (Project Hope)*, 30(7), 1335–1342. doi:10.1377/hlthaff.2011.0066
- Rust, G., Baltrus, P., Ye, J., Daniels, E., Quarshie, A., Boumbulian, P., & Strothers, H. (2009). Presence of a community health center and uninsured emergency department visit rates in rural counties. *The Journal of Rural Health: Official Journal of the American Rural Health Association and the National Rural Health Care Association*, 25(1), 8–16. doi:10.1111/j.1748-0361.2009.00193.x
- Schuur, J. D., & Venkatesh, A. K. (2012). The growing role of emergency departments in hospital admissions. *The New England Journal of Medicine*, 367(5), 391–393. doi:10.1056/NEJMp1204431
- Searing, A., Hoadley, J. (2016). Beyond the reduction in uncompensated care: Medicaid expansion is having a positive Impact on safety net hospitals and clinics. http://ccf.georgetown.edu/wp-content/uploads/2016/05/Medicaid_hospitals-clinics-June-2016.pdf
- Sharma, R., Lebrun-Harris, L. A., & Ngo-Metzger, Q. (2014). Costs and clinical quality among Medicare beneficiaries: associations with health center penetration of low-income residents. *Medicare & Medicaid Research Review*, 4(3), E1–E17. doi:10.5600/mmrr.004.03.a05
- Singh, J. V., & Lumsden, C. J. (1990). Theory and research in organizational ecology. *Annual Review of Sociology*, 16(1), 161–195. doi:10.1146/annurev.so.16.080190.001113
- Trzeciak, S., & Rivers, E. P. (2003). Emergency department overcrowding in the United States: an emerging threat to patient safety and public health. *Emergency Medicine Journal: EMJ*, 20(5), 402–405. doi:10.1136/emj.20.5.402
- Vickery, K. D., Sauser, K., & Davis, M. M. (2013). Policy responses to demand for health care access: from the individual to the population. *JAMA: The Journal of the American Medical Association*, 309(7), 665–666. doi:10.1001/jama.2012.96863

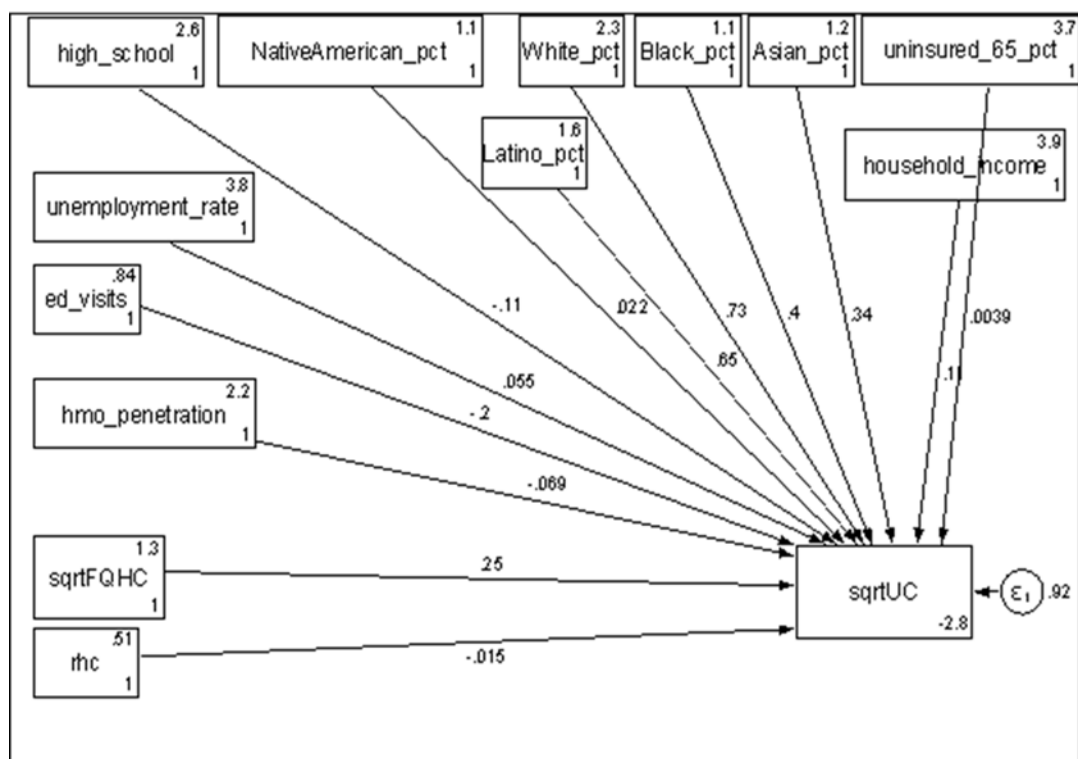
- Weiss, A., Long, S., Ramos, C., & Coughlin, T. (2016). *Federally Qualified Health Centers? Importance in the Safety Net Continues for Many as Affordable Care Act Implementation Moves Ahead*. Washington, DC: The Urban Institute.
- White, B. M., & Newman, S. D. (2015). Access to primary care services among the homeless: a synthesis of the literature using the equity of access to medical care framework. *Journal of Primary Care & Community Health*, 6(2), 77–87.
doi:10.1177/2150131914556122
- White, K. L., Williams, T. F., & Greenberg, B. G. (1961). The ecology of medical care. *The New England Journal of Medicine*, 265(18), 885–892.
doi:10.1056/NEJM196111022651805
- White, K. L. (1997). The ecology of medical care: origins and implications for population-based healthcare research. *Health Services Research*, 32(1), 11–21.
<https://www.ncbi.nlm.nih.gov/pubmed/9108800>

APPENDIX A
CORRELATION MATRIX

	physi cians	ed visit	hosp beds	hmo pen	medi caid	HS	unem plove d	unins ured	Hou inco me	rhc	sqrtF QHC	sqrtU C
Physi cians	1.00											
ED visits	0.97	1.00										
Hosp beds	0.98	0.99	1.00									
HM O penet ratio n	0.30	0.33	0.32	1.00								
Medi caid expa nsion	0.16	0.13	0.13	0.09	1.00							
High scho ol	0.15	0.19	0.19	0.02	-0.14	1.00						
Une mplo ymen t	0.08	0.12	0.11	0.03	0.08	0.46	1.00					
Unin sured 18+	0.01	0.08	0.07	-0.06	-0.63	0.56	0.16	1.00				
Hous ehold inco me	0.26	0.19	0.18	0.14	0.19	-0.43	-0.39	-0.35	1.00			
RHC	-0.13	-0.12	-0.12	-0.11	0.02	0.25	0.18	0.03	-0.22	1.00		
sqrtF QHC	0.87	0.87	0.87	0.37	0.21	0.18	0.22	0.00	0.21	-0.08	1.00	
sqrtU C	0.12	0.13	0.12	0.12	-0.09	0.01	0.03	0.08	0.09	-0.07	0.16	1.00

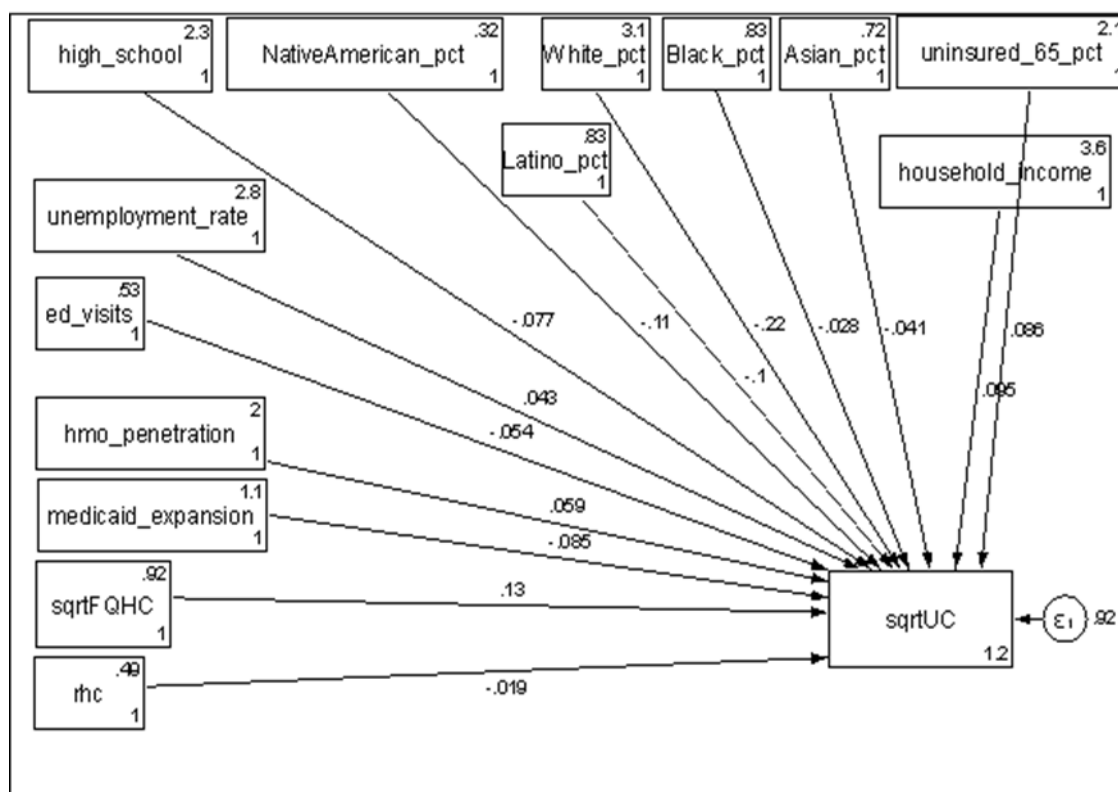
APPENDIX B

STRUCTURAL EQUATION MODELING (SEM)



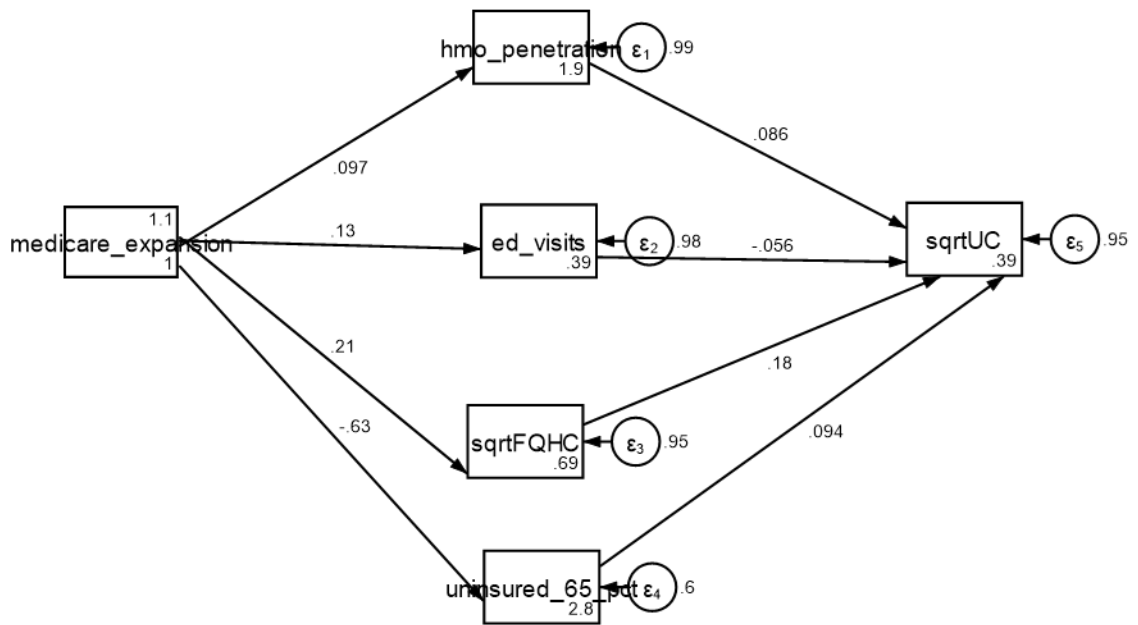
APPENDIX C

SEM WITH MEDICAID EXPANSION



APPENDIX D

SEM WITH MEDICAID EXPANSION AS MODERATOR



APPENDIX E
IRB APPROVAL

NHSR DETERMINATION

TO: Norris, Kena R

FROM: University of Alabama at Birmingham Institutional Review Board
Federalwide Assurance Number FWA00005960
IORG Registration # IRB00000196 (IRB 01)
IORG Registration # IRB00000726 (IRB 02)

DATE: 27-Dec-2018

RE: IRB-300002696
The Effect of Federally Qualified Health Centers (FQHCs) Prevalence on Uncompensated Care in U.S. Hospitals

The Office of the IRB has reviewed your Application for Not Human Subjects Research Designation for the above referenced project.

The reviewer has determined this project is not subject to FDA regulations and is not Human Subjects Research. Note that any changes to the project should be resubmitted to the Office of the IRB for determination.

if you have questions or concerns, please contact the Office of the IRB at 205-934-3789.